

Geography Ch-6 (Must-Do) — Landforms of Glaciation

Must-Do / Foundation+Standard | GS-I (Physical Geography) | Sources: GC Leong + Majid Husain

Glacial erosion & deposition: cirque, arete, horn, U-valley, moraine, drumlin, esker

CA anchor: Gangotri glacier retreat (Gomukh snout)

Facts from official sources | Inferences clearly marked | No fabricated data

HIGH

FOUNDATION — Glacial Erosion & Deposition Landforms (GC Leong / Majid)

GS-I • Physical
Geography →
Glacial Landforms
(Source: GC
Leong, Majid
Husain)

■ NEWS TRIGGER

Glacial landforms explain the high Himalaya's scenery and its water towers — the same glaciers now retreating (Gangotri) and spawning hazardous glacial lakes.

■ INTRO & ORIGIN

Moving ice erodes by **plucking** (freeze-thaw prising rock) and **abrasion** (scratching/grinding with embedded debris), then deposits its load as **moraines** and drift when it melts.

*A valley glacier fed by several corries flows downslope, gouges its floor and walls, and on melting drops unsorted **till** (moraine) and, via meltwater, **sorted outwash**.*

■ KEY DATA

Type	Landform	What it is
Erosional	Cirque (corrie/cwm)	Horse-shoe armchair basin at valley head; holds a tarn
Erosional	Arete / Horn	Knife-edge ridge / pyramidal peak (e.g. Matterhorn) between cirques
Erosional	U-shaped valley / Hanging valley	Glacial trough; tributary valley left 'hanging' above it
Erosional	Roche moutonnee / Fjord	Asymmetric rock bump / drowned glacial trough
Depositional	Moraines (lateral/medial/terminal/ground)	Ridges of unsorted till
Depositional	Drumlin / Esker / Outwash plain	Egg-shaped till mound / sinuous meltwater ridge / sorted sandur

■ STATIC THEORY

- **Cirque (corrie/cwm)**: plucking on the back-wall + abrasion of the floor deepen a horse-shoe basin; a small lake in it is a **tarn**.
- **Arete** = sharp ridge separating two cirques ('fish-bone'); where three or more cirques meet, a pyramidal **horn** forms (classic: Matterhorn).
- **U-shaped valley (glacial trough)**: the glacier grinds a former V-valley into a broad, steep-sided U; tributary valleys are left as **hanging valleys** (waterfalls).
- **Roche moutonnee**: asymmetric bump — gently abraded up-ice side, plucked steep down-ice side — a hallmark of glacial erosion; a drowned trough is a **fjord**.
- **Depositional (Majid)**: **moraines** (lateral, medial, terminal, ground) of unsorted till; **drumlins** (oval till mounds), **eskers** (meltwater ridges), and **outwash plains/sandur** of sorted sediment.

■ MUST-KNOW FACTS

1. Glacial erosion = plucking + abrasion.

■ UPSC TRAPS

- ✗ U-shaped valleys are made by rivers.

2. Cirque -> tarn; arete = ridge; horn = pyramidal peak (Matterhorn). 3. U-shaped valley + hanging valleys are glacial (rivers make V-valleys). 4. Roche moutonnee: gentle up-ice, steep plucked down-ice. 5. Moraine = unsorted till; outwash plain (sandur) = sorted meltwater deposit. 6. Drumlin (till mound) vs esker (meltwater ridge).	✓ Rivers make V-shaped valleys; glaciers carve U-shaped troughs. ✗ Moraine and outwash are both sorted. ✓ Moraine (till) is UNSORTED; outwash (meltwater) is SORTED. ✗ A tarn is a depositional landform. ✓ A tarn is a lake in an EROSIONAL cirque basin.
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■ MAINS ANGLE

Distinguish glacial erosional vs depositional landforms and use them to read the physiography and water-tower role of the high Himalaya.

■ STUDY LINK

Geography → Physical → Glacial Landforms
(GC Leong: Ice; Majid: Glacial Geomorphology)

■ MEDIUM UPSC RELEVANCE

MEDIUM

CA ANCHOR — Gangotri Glacier Retreat (Gomukh Snout)

GS-I / GS-III ·
Glaciation applied
→ Climate
change & rivers

■ NEWS TRIGGER

The Gangotri glacier — source of the Ganga at the Gomukh snout — has retreated ~3.1 km between 1936 and 2020, a marker of Himalayan warming and a threat to perennial river flow.

■ INTRO & ORIGIN

A retreating glacier snout is the visible edge of a shrinking cirque/valley-glacier system — directly linking glacial geomorphology to climate change and water security.

■ STATIC THEORY

- The glacier terminus is the **snout** (here **Gomukh**), from which the Bhagirathi (headstream of the Ganga) emerges.
- Retreat exposes fresh moraine and pro-glacial ground; the retreat rate has recently slowed (~15-20 m/yr) but continues.
- Impact: long-term risk to **perennial river flow**, and formation of unstable glacial lakes upstream.

■ MUST-KNOW FACTS

1. Gangotri snout = Gomukh; source of Bhagirathi -> Ganga.
2. Retreat ~3.1 km (1936-2020); ongoing due to warming.
3. Glacier retreat = shrinking valley/cirque glacier (geomorphology + climate).

■ UPSC TRAPS

- ✗ The Ganga rises at Gangotri town.
- ✓ The Bhagirathi emerges from the GOMUKH snout of the Gangotri glacier.

■ MAINS ANGLE

Connect glacial geomorphology to climate-driven retreat and India's water-security stakes.

■ STUDY LINK

Geography → Glaciation → Glacier Retreat (applied CA)

End of Register Notes — UPSC Agent / Copilot CLI